



Solar & Battery Room Monitor Documentation

Low-voltage monitoring demo for solar, battery, generator, and mobile trailer environments

Rev 2 - Trailer Battery and EcoFlow Monitoring Update

CELESTIAL ELECTRIC

SOLAR & BATTERY ROOM MONITOR

LOW-VOLTAGE MONITORING. REAL-TIME VISIBILITY. PEACE OF MIND.

- TEMPERATURE MONITORING
- HUMIDITY MONITORING
- WATER LEAK DETECTION
- DOOR / CONTACT STATUS
- CLOUD DASHBOARD & ALERTS

TEMPERATURE
72.4 °F

HUMIDITY
45 %

WATER LEAK
DRY
No water detected

DOOR / CONTACT
CLOSED
Door is closed

STATUS
All systems normal
Last updated: 11:32 AM

LOW-VOLTAGE MONITORING ONLY | ARDUINO CLOUD CONNECTED | REAL-TIME ALERTS & NOTIFICATIONS | BUILD BETTER. MONITOR SMARTER.

Prepared for Arduino Project Hub public documentation and downloadable project files.

Project Overview

This project demonstrates a simple low-voltage monitoring system for solar PV, battery storage, generator, trailer, and electrical equipment spaces. The system uses an Arduino UNO R4 WiFi, environmental sensing, water/leak detection, a door/contact input, local status LEDs, and Arduino Cloud dashboard visibility.

The project is designed for training, public demonstration, field-service concepts, and customer education. It shows how low-voltage sensors, field wiring, dashboards, and documentation can work together to provide better visibility into critical electrical spaces.

Monitored Equipment and Use Cases

Use Case	Monitoring Goal
Solar/battery equipment area	Temperature, humidity, water/leak status, access/contact status, and local normal/alert indication.
Generator or backup-power room	Environmental and contact status monitoring only. Generator/ATS control is outside the scope of this demo.
Mobile service trailer	Monitor the 12V Duracell battery bank area, EcoFlow power area, water/leak status, and compartment access/contact status.
Training lab	Teach low-voltage wiring, sensor inputs, dashboard variables, safety boundaries, and documentation discipline.

Trailer Battery and EcoFlow System

Item	Project Detail
Primary monitored 12V bank	Two Duracell 12V lead-acid batteries, each rated 810 cold cranking amps (CCA), installed in parallel.
Parallel wiring meaning	The batteries remain a 12V nominal battery bank. Available starting current and reserve capacity are increased when the batteries are properly matched and wired.
Nominal combined CCA	Two matched 810 CCA batteries in parallel are nominally capable of about 1,620 CCA. This is a starting-current rating, not a usable energy-capacity rating.
Monitoring target	Bank voltage, compartment temperature/humidity, water/leak status, door/contact status, local normal/alert LED indication, and cloud dashboard visibility.
EcoFlow system	The trailer also includes an EcoFlow portable power system. The exact EcoFlow model, capacity, and solar/AC input ratings should be confirmed from the unit nameplate and added to the final documentation.
Trailer role	Celestial Electric mobile service, demonstration, and training trailer used to show low-voltage monitoring, field diagnostics, and mobile energy concepts.

The two Duracell batteries are documented as a 12V parallel bank. The Arduino demonstration may monitor bank voltage through a properly fused low-current sense input, plus environmental conditions around the battery/power compartment. The Arduino must not carry battery load current, starter current, inverter current, or charger current.

Suggested Monitoring Architecture

Layer	Description
Sensors	Temperature/humidity sensor, magnetic contact sensor, water/leak sensor, optional fused battery-voltage sense input, and local red/green status LEDs.
Controller	Arduino UNO R4 WiFi reads low-voltage inputs and updates Arduino Cloud variables.
Dashboard	Arduino Cloud dashboard displays normal/alert status, environment values, water/leak status, contact status, and optional trailer/battery status.
Documentation	Project Hub writeup, safety note, BOM, build checklist, and trailer battery monitoring overview.

Dashboard Variables

Dashboard Variable	Purpose
batteryBankVoltage	Displays the approximate 12V Duracell battery bank voltage using a fused voltage sense input.
batteryCompartmentTemp	Tracks compartment temperature near the battery or power equipment area.
batteryCompartmentHumidity	Tracks humidity in the monitored compartment.
waterLeakDetected	Shows whether water is detected near the equipment compartment or trailer power area.
doorContactStatus	Shows whether the compartment door, access door, or enclosure contact is open or closed.
ecoFlowStatus	Optional manual or future integration field for EcoFlow system status.
systemStatus	Simple summary: Normal, Attention Required, or Alert.

Components Summary

Component	Purpose
Arduino UNO R4 WiFi	Main Wi-Fi enabled controller for sensor inputs and Arduino Cloud dashboard communication.
BME280 or DHT20 sensor	Temperature and humidity monitoring. BME280 is preferred for the professional version.
Magnetic contact switch	Door, enclosure, or compartment access status.
Water leak detection sensor	Water or moisture detection near equipment area.
Red and green LEDs with resistors	Local normal/alert indication.
Breadboard, wiring, enclosure, USB power	Prototype and demo assembly.
Fused voltage-sense input for 12V bank	Optional battery-bank voltage monitoring. This must be low-current and protected at the battery source.

Build Steps

- Mount the Arduino, sensors, and indicators in a clean prototype enclosure or training board.
- Connect the temperature/humidity sensor using I2C or the required digital sensor wiring.
- Wire the magnetic contact as a dry-contact input using a pull-up or pull-down method.
- Wire the water/leak sensor to a low-voltage digital input.
- Wire red and green LEDs through proper current-limiting resistors.
- If battery-bank voltage is monitored, use fused low-current sensing and proper voltage scaling/isolation before connecting to any Arduino input.
- Create Arduino Cloud variables and dashboard widgets for temperature, humidity, leak status, contact status, battery voltage, and system status.

- Test each sensor in a safe low-voltage bench environment before using the trailer as a demonstration platform.

Safety Note

This project is for low-voltage monitoring, training, and demonstration only. Do not connect an Arduino directly to service equipment, PV strings, battery terminals, inverter terminals, generator controls, transfer switches, utility equipment, fire alarm circuits, or line-voltage wiring. Any real field or trailer installation must use proper listed equipment, isolation, overcurrent protection, enclosures, strain relief, wire separation, and code-compliant wiring methods.

This project does not replace listed safety controls, manufacturer monitoring systems, utility-required equipment, fire alarm systems, generator/ATS controls, battery management systems, EcoFlow manufacturer controls, or code-required protective devices.

Publication Notes

For Arduino Project Hub, attach this documentation as the main public PDF. Use the separate Trailer Battery & EcoFlow Monitoring Overview PDF as a downloadable support file. The Project Hub Upload Guide is an internal checklist and does not need to be published as a public downloadable file.